

Ayush Girhay

B.Tech. III Year, Civil Engineering
Indian Institute of Technology Roorkee

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Profile

AI/ML, Large Language Models, NLP, Data Science, Deep Learning, Web-Development, Data Structures and Algorithms

AIR-6707 JEE Advanced 2023 **AIR-1283** JEE Mains 2023

Education

Indian Institute of Technology, Roorkee 2023 – Present
B.Tech., Civil Engineering

Insight Junior College (MSBSHSE) 2021 – 2022
Intermediate (Class XII)

Experience

TRS Consultant May 2026 – Present
Worked as a **Consultant Intern**, utilizing business software to support daily operations, manage client-related tasks, maintain records, and improve workflow efficiency.

Curious About Sales FEB 2026 - APR 2026
Worked as an **AI Engineer Intern**, building AI-powered solutions with Python and Generative AI, including automation workflows, prompt engineering, and AI prototypes to improve operational efficiency.

Projects

Creating a Large Language Model Jun 2026 – Present
Built a decoder-only LLM from scratch in Python by implementing tokenization, embeddings, transformer blocks, self-attention, and autoregressive text generation with a complete training pipeline.

EventSphere – AI-Powered Event and Media Management Platform May 2026 – Jun 2026
Developed a full-stack event management platform using React, Node.js, Express, and MongoDB featuring authentication, ticket booking, QR attendance, media sharing, and role-based access control.

Dynamic Tariff Optimization for EV Charging Networks Sep 2023 – Nov 2023
Built an AI-driven pricing engine for EV charging stations using demand forecasting and occupancy prediction to optimize tariffs and improve resource utilization.

Dynamic Pricing for Urban Parking Lots May 2025 – Jul 2025
Consulting & Analytics Club IIT Guwahati x Pathway
Developed a real-time parking pricing system using Python, Pandas, NumPy, Pathway, and Bokeh by analyzing demand, traffic, occupancy, and competitor pricing across 14 parking lots.

House Price Prediction Dec 2025 – Jan 2026
Developed a multimodal ML model combining tabular data and satellite imagery using XGBoost, PCA, SHAP, and Grad-CAM for accurate and explainable house price prediction.

Technical Skills

Languages: Python, C++, SQL, HTML, CSS, JavaScript, Bootstrap, React.js, ROS2

Data Analytics: EDA, Pandas, NumPy, Matplotlib, Data Visualization

Software Packages: ArcGIS, QGIS, Solidworks, Fusion360

Tools: Jupyter Notebook, Google Colab, scikit-learn, MySQL, Tableau, Excel, Git, GitHub, VS Code, Postman

Achievements

Silver Medalist – 58th Inter IIT Sports Meet (Hockey)

Institute Excellence Award – For Outstanding Overall Performance

Codeforces – Maximum Rating: **1500+**

Positions of Responsibility

Captain — Hockey, IIT Roorkee Jul 2026 – Present
Led the IIT Roorkee hockey team, coordinating training, mentoring players, and representing the institute in tournaments while strengthening leadership, teamwork, and strategic decision-making.

Core Member — Models and Robotics Section (MARS), IIT Roorkee Jun 2025 – Present
Contribute to robotics projects involving CAD design, AI/ML integration, and autonomous systems using SolidWorks and Fusion 360.